

Propiedades de la convolución

Lema 1. Sean $f, g, h: \mathbb{R}^n \rightarrow \mathbb{K} = \mathbb{R} \text{ o } \mathbb{C}$ dos funciones medibles, entonces

- (1) Conmutatividad: $f * g = g * f$ c.t.p.
- (2) Asociatividad: $f * (g * h) = (f * g) * h$ c.t.p.
- (3) $T_z(f * g) = (T_z f) * g = f * (T_z g)$ c.t.p.
- (4) $\text{supp}(f * g) \subset \text{supp}(f) + \text{supp}(g)$.

Demostración:

$$(1) \quad f * g(x) = \int_{\mathbb{R}^n} f(x-y) g(y) dy \stackrel{z=x-y}{=} \int_{\mathbb{R}^n} f(z) g(x-z) dz = g * f(x) \text{ c.t.p. } x \in \mathbb{R}^n.$$

(2) Calculamos y aplicamos el Teorema de Fubini:

$$\begin{aligned} (f * g) * h(x) &= \int_{\mathbb{R}^n} (f * g)(x-y) h(y) dy = \int_{\mathbb{R}^n} g * f(x-y) h(y) dy \\ &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} g(x-y-z) f(z) dz \right) h(y) dy \\ &\stackrel{\text{Fubini}}{=} \int_{\mathbb{R}^n} f(z) \left(\int_{\mathbb{R}^n} g(x-y-z) h(y) dy \right) dz = \int_{\mathbb{R}^n} g * h(x-z) f(z) dz \\ &= f * (g * h)(x) \end{aligned}$$

(3) Se demuestra la primera igualdad, la segunda es análoga. Tenemos que

$$\begin{aligned} T_z(f * g)(x) &= f * g(x-z) = \int_{\mathbb{R}^n} f(x-z-y) g(y) dy \\ &= \int_{\mathbb{R}^n} T_z f(x-y) g(y) dy = (T_z f) * g(x). \end{aligned}$$

(4) Basta probarlo para el soporte sin cerrar. Es decir, si $s(f) := \{x \in \mathbb{R}^n : f(x) \neq 0\}$ basta ver que $s(f * g) \subset s(f) + s(g)$. Sea $x \in \mathbb{R}^n \setminus (s(f) + s(g))$, entonces $\forall y \in s(g) : f(x-y) = 0$ porque $f(x-y) \neq 0 \implies x-y \in s(f) \implies x = (x-y) + y \in s(f) + s(g)$.

Como $f(x-y) = 0$, $f * g(x) = \int_{\mathbb{R}^n} f(x-y) g(y) dy = 0$, luego $x \in \mathbb{R}^n \setminus s(f * g)$. ■

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